# Linguistic annotations

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# Topics to cover

- Review (10 mins)
- Presentation (20 mins)
- Linguistic annotations (30 mins)
  - What
  - Why
  - How
- Lab: Annotation exercise

# Review:
# Measuring constructs in SLA

# Learner corpus research - SLA

## Goal

To turn learner language into empirical evidence about SLA.

- Learner corpora contain observable language production.
- SLA research asks questions about *theoretical* constructs.

# Learner corpus research - SLA

Complexity, accuracy, fluency are well-known theoretical constructs in SLA.

To investigate them in corpora, we need to decide:

- What exactly does the construct mean?
  *operationalization*

- Which linguistic features represent it?
  e.g., syntactic hierarchies (words in clauses), grammatical categories (noun - noun modifiers)

- How will those features be measured?
  e.g., length (mean length of X), diversity (different types of grammatical structures in X)

# Development involves more than accuracy

The complexity–accuracy–fluency (CAF) framework distinguishes several aspects of learner performance (Norris & Oretega, 2009).

| | |
|---|---|
| Accuracy | Conformity to target-language norms. |
| Fluency | Efficiency and automatization of language use. |
| **Complexity** | **Expansion and elaboration of linguistic resources.** |

# Complexity

- Syntactic complexity (some examples in last week's presentation)
- Grammatical complexity (today's presentation)

# Complexity: Examples

- A: The students very carefully reviewed the report yesterday afternoon.
- B: The students reviewed the report on local education policy.
- Both: 9 words, 1 clause $\rightarrow$ ?
- Grammatical structures
  - A: ?
  - B: ?
- Syntactic and grammatical complexity measures may reveal different aspects of learner production (with patterns varying across *spoken* and *written* modes and across *registers*).

# Presentation: Biber & Larsson (2026)

# Linguistic annotations: What

# What is linguistic annotation

- Corpus analysis often involves more than raw text.
- We add linguistic information to identify and analyze patterns.
- Linguistic annotation = labeling language data with information.

Fruit flies like an apple.

Fruit flies like an apple.

Time flies like an arrow.

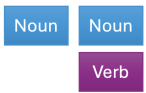
Fruit flies like an apple.

Noun

Time flies like an arrow.

Noun

Fruit flies like an apple.




Time flies like an arrow.



Fruit flies like an apple.




Time flies like an arrow.




Fruit flies like an apple.
Noun
Noun
Prep
Det
Noun
Verb
Verb


Time flies like an arrow.
Noun
Noun
Prep
Det
Noun
Verb
Verb







Make a decision based on linguistic interpretation of the given dataset

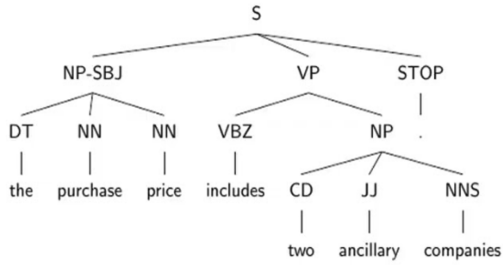
# Different types of linguistic annotations

- **Type: Syntactic information**

- *English Penn Treebank corpus*
  - Used for character-level and word-level *language modeling*

**Table 2**
The Penn Treebank POS tagset.

| | | | |
|---|---|---|---|
| 1. | CC | Coordinating conjunction | |
| 2. | CD | Cardinal number | |
| 3. | DT | Determiner | |
| 4. | EX | Existential *there* | |
| 5. | FW | Foreign word | |
| 6. | IN | Preposition/subordinating conjunction | |
| 7. | JJ | Adjective | |
| 8. | JJR | Adjective, comparative | |
| 9. | JJS | Adjective, superlative | |
| 10. | LS | List item marker | |
| 11. | MD | Modal | |
| 12. | NN | Noun, singular or mass | |
| 13. | NNS | Noun, plural | |
| 14. | NNP | Proper noun, singular | |
| 15. | NNPS | Proper noun, plural | |
| 16. | PDT | Predeterminer | |
| 17. | POS | Possessive ending | |
| 18. | PRP | Personal pronoun | |
| 19. | PP$ | Possessive pronoun | |
| 20. | RB | Adverb | |
| 21. | RBR | Adverb, comparative | |
| 22. | RBS | Adverb, superlative | |
| 23. | RP | Particle | |
| 24. | SYM | Symbol (mathematical or scientific) | |
| 25. | TO | *to* | |
| 26. | UH | Interjection | |
| 27. | VB | Verb, base form | |
| 28. | VBD | Verb, past tense | |
| 29. | VBG | Verb, gerund/present participle | |
| 30. | VBN | Verb, past participle | |
| 31. | VBP | Verb, non-3rd ps. sing. present | |
| 32. | VBZ | Verb, 3rd ps. sing. present | |
| 33. | WDT | *wh*-determiner | |
| 34. | WP | *wh*-pronoun | |
| 35. | WP$ | Possessive *wh*-pronoun | |
| 36. | WRB | *wh*-adverb | |
| 37. | # | Pound sign | |
| 38. | $ | Dollar sign | |
| 39. | . | Sentence-final punctuation | |
| 40. | , | Comma | |
| 41. | : | Colon, semi-colon | |
| 42. | ( | Left bracket character | |
| 43. | ) | Right bracket character | |
| 44. | " | Straight double quote | |
| 45. | ‘ | Left open single quote | |
| 46. | “ | Left open double quote | |
| 47. | ’ | Right close single quote | |
| 48. | ” | Right close double quote | |




Marcus, M., Santorini, B., & Marcinkiewicz, M. A. (1993). Building a large annotated corpus of English: The Penn Treebank. *Computational Linguistics*, *19*(2), 313–330.

# More examples of syntactic annotation resources

| Resource | Annotation |
|---|---|
| UD Treebank | Shared POS tags, morphological features, and dependency relations |
| Prague Treebank | Originally annotated for Czech using the Prague framework (project begun in 1996); available in converted UD form |
| KAIST Treebank | Originally annotated with constituency structures for Korean (described in Choi et al., 1994); available in converted UD form |

- Hajič, J., & Hajičová, E. (1997). Syntactic tagging in the Prague Treebank.
- Choi, K.-S., Han, Y. S., Han, Y. G., & Kwon, O. W. (1994). KAIST Tree Bank Project for Korean: Present and Future Development. In *Proceedings of the International Workshop on Sharable Natural Language Resources* (pp. 7–14).

# More and more examples (of different annotations)

Table 9: Representative types of linguistic annotation.

| Type of annotation | Description | Examples |
|---|---|---|
| POS tagging | Adding information about the grammatical category (part of speech) of each word in a text. | Noun, verb, adjective, adverb. |
| Phonetic annotation | Adding information about how words or sounds are pronounced in a spoken corpus. | IPA transcription, individual speech sounds. |
| Prosodic annotation | Adding information about the prosodic features of spoken language. | Stress, intonation, pauses, rhythm. |
| Syntactic annotation | Adding information about the syntactic structure of sentences, including how words are organized into phrases and clauses. | Constituency parsing, dependency parsing. |
| Semantic annotation | Adding information about the meanings or semantic categories of words and expressions. | *Cricket* as a sport vs. *cricket* as an insect. |

Leech (2004). Adding Linguistic Annotation. Section 2

# More and more examples (of different annotations)

| | | |
|---|---|---|
| Discourse annotation | Adding information about relationships between linguistic elements across sentences or larger stretches of text. | Anaphoric links between pronouns and their antecedents. |
| Stylistic annotation | Adding information about stylistic features, including how speech and thought are represented in a text. | Direct speech, indirect speech, free indirect thought. |
| Lexical annotation | Adding information about the lexical identity of words, such as the lemma associated with each word form. | *Running*, *runs*, *ran* $\rightarrow$ *run*. |
| Error annotation | Adding information about errors in language production, particularly in learner corpora, including their locations and types. | Grammatical errors, lexical errors, spelling errors, word-order errors. |

Leech (2004). Adding Linguistic Annotation. Section 2

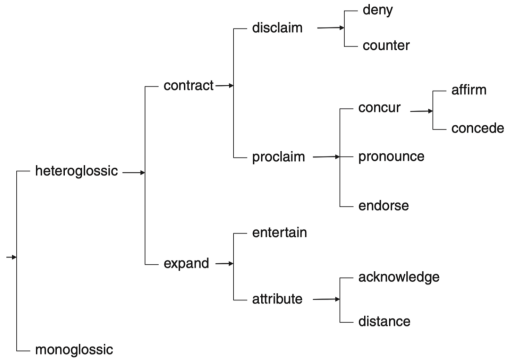
# Reinventing the wheel

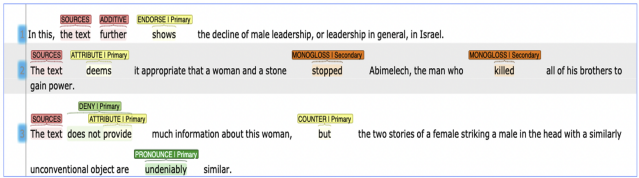
- What if existing annotation schemes do not meet our needs?
- Many existing schemes annotate individual tokens or relations between them. Recall POS tags, dependency structures:
  - POS tags: Grammatical categories assigned to tokens
  - Dependency structures: Syntactic relations between tokens
- What if our unit of analysis extends beyond individual words?

# MORE examples :)

**Target linguistic structures (beyond word-level)**

- Stance-taking (Engagement) movement in writing (Eguchi & Kyle, 2023)



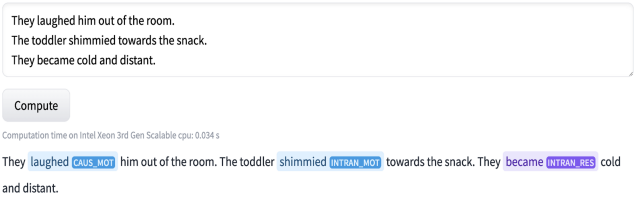



# MORE examples :)

## Target linguistic structures (beyond word-level)

- Stance-taking (Engagement) movement in writing (Eguchi & Kyle, 2023)
- Argument structure constructions (Kyle & Sung, 2023; Sung & Kyle, 2024)

| ASC tag | Syntactic frame (example) | Semantic frame (example) |
|---|---|---|
| ATTR | Subj-$V_{copular}$-NP/AdjP/PP | $X_{theme}$ is $Y_{attribute}$ |
| INTRAN_S | Subj-V | $X_{agent}$ acts |
| INTRAN_MOT | Subj-V-Adverbial Particle/PP | $X_{theme}$ moves $Y_{path}$ |
| INTRAN_RES | Subj-V-NP/AdjP | $X_{patient}$ becomes $Y_{result}$ |
| TRAN_S | Subj-V-Obj | $X_{agent}$ acts on $Y_{patient}$ |
| DITRAN | Subj-V-$Obj_{indirect}$-$Obj_{direct}$ | $X_{agent}$ causes $Y_{recipient}$ to receive $Z_{theme}$ |
| CAUS_MOT | Subj-V-Obj-PP | $X_{agent}$ causes $Y_{theme}$ to move $Z_{path/goal}$ |
| TRAN_RES | Subj-V-Obj-NP/AdjP | $X_{agent}$ causes $Y_{theme}$ to become $Z_{state}$ |
| PASSIVE | Subj-aux$V_{past\ participle}$ (*-by*-PP) | $X_{theme}$ undergo V (*by* $Y_{agent}$) |



# MORE examples :)

## Target linguistic structures (beyond word-level)

- Stance-taking (Engagement) movement in writing (Eguchi & Kyle, 2023)
- Argument structure constructions (Kyle & Sung, 2023; Sung & Kyle, 2024)
- Lexicogrammatical features (Kyle et al., *in prep*)

| idx | word | lemma | cx tag | tag | cat1 | cat2 | cat3 | cat4 | cat5 | cat6 | cat7 | cat8 | xpos | upos | deprel | headid |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | This | this | | pro | dem | sg | | | | | | | DT | PRON | nsubj | 1 |
| 1 | is | be | | vbmain | be | pres | simple | active | | | | | VBZ | AUX | ROOT | 1 |
| 2 | a | a | | dt | art | | | | | | | | DT | DET | det | 5 |
| 3 | very | very | rb+jjrbmod | rb | adjmod | | | | | | | | RB | ADV | advmod | 4 |
| 4 | important | important | attr+npremod | jj | attr | | | | | | | | JJ | ADJ | amod | 5 |
| 5 | opportunity | opportunity | | nn | | nom | | | | | | | NN | NOUN | attr | 1 |
| 6 | that | that | | relpro | relpro_that | sg | | | | | | | WDT | PRON | nsubj | 8 |
| 7 | only | only | rb+advl | rb | advl | ly | | | | | | | RB | ADV | advmod | 8 |
| 8 | comes | come | finitecls+rel | vbmain | vblex | pres | simple | active | nmod_cls | thatcls | rel | | VBZ | VERB | relcl | 5 |
| 9 | once | once | rb+advl | rb | advl | | | | | | | | RB | ADV | advmod | 8 |
| 10 | in | in | in+advl | in | advl | | | | | | | | IN | ADP | prep | 8 |
| 11 | a | a | | dt | art | | | | | | | | DT | DET | det | 12 |
| 12 | lifetime | lifetime | | nn | | | | | | | | | NN | NOUN | pobj | 10 |
| 13 | . | . | | | | | | | | | | | . | PUNCT | punct | 1 |

# Linguistic annotations: Why

# Benefits of linguistic annotation

Beyond the sheer joy of annotating…

# Benefits of linguistic annotation

- Efficient retrieval: Search for linguistic categories and structures beyond word forms.
  - They *separate* the groups. → Verb
  - They live in *separate* houses. → Adjective
  - Dependency annotation allows searches for subjects or objects of particular verbs.

# Benefits of linguistic annotation

- Reusability: Annotate once, investigate many questions.
  - Annotation requires substantial time and effort, including manual review of automatic output.
  - Shared resources reduce duplicated effort and support comparisons across studies; open science

# Benefits of linguistic annotation

- cf. data labeling: Assigning labels to data, often to train and evaluate machine learning models

- Provide labeled data for language-related machine learning tasks

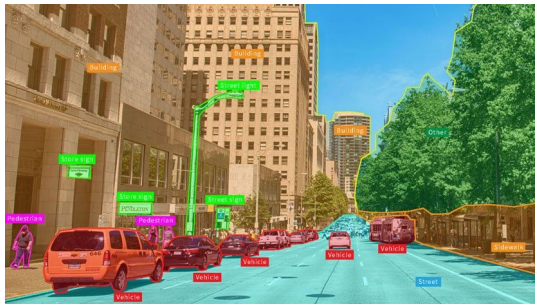



Image source from `https://kr.appen.com/blog/data-annotation/`

# Principles of good annotation practice

- Preserve the original data
  - Keep the original text or recording recoverable.
  - Use inline or stand-off annotation with traceable links to the source.
  - Document normalization: corrections must not silently replace what speakers or writers produced!
- Document the annotation process
  - Specify the scheme, tools and model versions, annotators, and annotation stages.
  - Record training, quality checks, revisions, and disagreement resolution.

# Linguistic annotations: How

# Evaluating annotation

- **Validity**: Does the scheme capture meaningful linguistic distinctions?
- **Accuracy**: Do assigned labels match the reference labels?
- **Consistency**: Do independent annotators apply the scheme in the same way?

# Validity

- Consistent labeling != a linguistically meaningful annotation scheme.

# Validity

- Human interpretation: Annotators may interpret the same example differently.
  - We often recruit undergraduate linguistics majors (and/or crowdsourcing).
  - Even with linguistic training, annotators may disagree.
- Unclear boundaries: Linguistic categories are not always clear-cut.

# Example: *Kick the Bucket*

- **Task**: Label constructions as `transitive` or `intransitive`.
- **Case 1**: *kick the ball*
  - Verb + direct object $\rightarrow$ `transitive`
- **Case 2**: *kick the bucket*
  - Literal reading: Physically kick a bucket.
  - Idiomatic reading: Die.
  - Should the interpretation affect our label?
- What are we annotating: **syntactic form**, **meaning**, or both?

# Example: *Kick the Bucket*

- **Option 1: Annotate syntactic form**
  - Label both expressions as `transitive`: both have a verb + direct object structure.
- **Option 2: Annotate form and idiomaticity separately**
  - *kick the ball*: `transitive`, `non-idiomatic`
  - *kick the bucket* (meaning “die”): `transitive`, `idiomatic`
  - Idiomaticity and transitivity are different dimensions.
- **Document the decision**: Define the categories, provide examples, and explain difficult cases (from the informed theory/framework)

# Accuracy

- Compare assigned labels with a reference or “gold” standard.
- Accuracy: The proportion of evaluated units labeled correctly.

$$A = \frac{C}{N}$$

- $C$: Number of correctly labeled units
- $N$: Total number of evaluated units
- Accuracy score can hide poor performance on particular categories.
- Precision and recall help identify different types of error.

# Precision/Recall: Identifying nouns

- Suppose `NOUN` is the target category.

| | **Gold: NOUN** | **Gold: Not NOUN** |
|---|---|---|
| **Predicted: NOUN** | True positive (TP) | False positive (FP) |
| **Predicted: Not NOUN** | False negative (FN) | True negative (TN) |

- **Precision**: Of the tokens labeled as nouns, how many really are nouns?

$$P = \frac{TP}{TP + FP}$$

- **Recall**: Of all actual nouns, how many did the model identify?

$$R = \frac{TP}{TP + FN}$$

# Worked Example: Two Noun Taggers

- A corpus contains 100 actual noun tokens.

| | **Model A** | **Model B** |
|---|---|---|
| Tokens labeled as nouns | 50 | 150 |
| Correctly identified nouns (TP) | 48 | 95 |
| Incorrect noun predictions (FP) | 2 | 55 |
| Missed nouns (FN) | 52 | 5 |
| Precision | $48/50 = .96$ | $95/150 \approx .63$ |
| Recall | $48/100 = .48$ | $95/100 = .95$ |

- Model A: Few false positives, but many missed nouns.
- Model B: Finds most nouns, but includes many non-nouns.
- Which model better serves your research question?

# F1 and Performance Across Categories

- F1: The harmonic mean of precision and recall.

$$F_1 = \frac{2PR}{P + R}$$

  - Model A: $F_1 = .64$
  - Model B: $F_1 = .76$

- For multiple categories, report scores by label and specify how overall scores are calculated.
  - *Micro*-F1: Pool counts across categories.
  - *Macro*-F1: Calculate scores for each category, then average them equally.

# Consistency

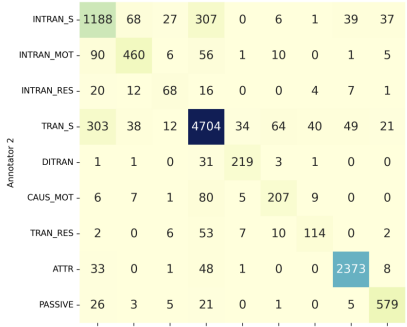
- Do independent human annotators assign the same labels?
- Simple agreement: The proportion of decisions with matching labels.
- Cohen's $\kappa$: Adjusts observed agreement for agreement expected from category frequencies.

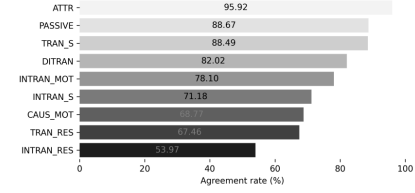
$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

  - $P_o$: Observed agreement
  - $P_e$: Expected agreement under the chance model

# Consistency

- Use a confusion matrix to identify categories that annotators confuse.




Sung & Kyle (2024). Annotation Scheme for English ASC Treebank.

# Lab: Lexicogrammatical feature annotation

# Annotation workshop: Goals and tasks

- **Goal**: Annotate four lexicogrammatical features in sample data and develop reusable annotation guidelines.

- **Task 1: Annotate and develop guidelines (20 minutes)**
  - Individually annotate sentences for your assigned feature.
  - Compare annotations within your group and discuss disagreements.
  - Write guidelines with definitions, examples, and rules for difficult cases.

- **Task 2: Exchange and test guidelines (20 minutes)**
  - Read the guidelines from your paired group.
  - Annotate sentences for the new feature only using their guidelines.
  - Compare annotations across groups and revise the guidelines.

- Resources here: `https://hksung.github.io/PythonTutorial/8.html`

# Annotation Workshop: Group Assignments

| Group | Task 1 | Task 2 |
|---|---|---|
| Group 1 | Verb + *that*-complement clause | Noun + *that*-complement clause |
| Group 2 | Noun + *that*-complement clause | Verb + *that*-complement clause |
| Group 3 | *-ing* clause as adverbial | Noun + *-ing* relative clause |
| Group 4 | Noun + *-ing* relative clause | *-ing* clause as adverbial |
| Group 5 | Verb + *-to* complement clause | Adjective + *-to* complement clause |
| Group 6 | Adjective + *-to* complement clause | Verb + *-to* complement clause |

- **Paired groups**: Groups 1 and 2; Groups 3 and 4; Groups 5 and 6

# Annotation Exercise: Group Debrief

- What worked well, and what challenges did your group encounter?
- Which instructions were helpful? Which were unclear or missing?
- What additional resources or training would have helped?
- What would you change to improve future annotation work?
- Any other thoughts or observations?

# Assignment 3: Linguistic Annotation

- Submit your annotation files (.txt format) for Task 1 and Task 2.
- Each file is worth 1 points (2 points total).
- Make sure that you include your outputs from the POS/dep parsing.
- Due this Friday!